

9.2 General Fourier Series and Convergence

we've been assuming $f(x)$ has period 2π and defined on $-\pi < x < \pi$ and the resulting Fourier series is

$$f(x) \sim \frac{1}{2}a_0 + \sum_{n=1}^{\infty} (a_n \cos(nx) + b_n \sin(nx))$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx$$

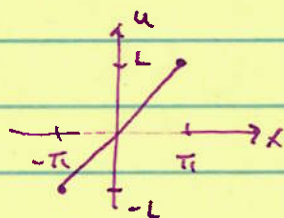
now let's first relax the period 2π part

let the period be $2L$ (where L is the half-period) and

have the function given on $-L < x < L$

redefine the variable: $u = \frac{L}{\pi} x$ this is so when $x = -\pi$ $u = -L$

$x = \pi$ $u = L$



replace x with $x = \frac{u\pi}{L}$ in the formulas above

$$f(u) \sim \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n \cos\left(n \cdot \frac{u\pi}{L}\right) + b_n \sin\left(n \cdot \frac{u\pi}{L}\right)$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx \text{ becomes}$$

$$= \frac{1}{\pi} \int_{-L}^L f(u) \cos\left(\frac{n\pi u}{L}\right) \cdot \frac{\pi}{L} du = \frac{1}{L} \int_{-L}^L f(u) \cos\left(\frac{n\pi u}{L}\right) du$$

$$\text{likewise, } b_n = \frac{1}{L} \int_{-L}^L f(u) \sin\left(\frac{n\pi u}{L}\right) du$$

but u is just a variable, we can call it whatever we want
(x or t or whatever)

now we have:

$f(x)$ periodic w/ period $2L$

given on $-L < x < L$

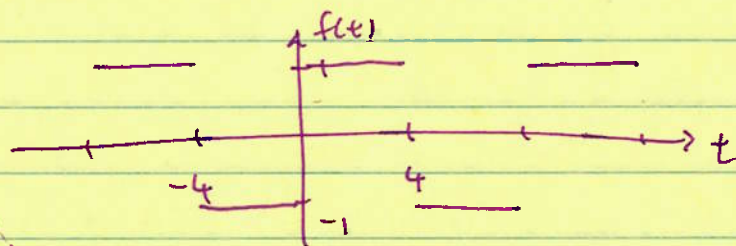
$$f(x) \sim \frac{1}{2}a_0 + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right]$$

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx$$

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

example $f(t) = \begin{cases} -1 & -4 < t < 0 \\ 1 & 0 < t < 4 \end{cases}$

period 8 ($2L=8$ so $L=4$)
half period



$a_0 = \frac{1}{L} \int_{-L}^L f(t) dt$
 $= \frac{1}{4} (0) = 0$
 net area

$$a_n = \frac{1}{L} \int_{-L}^L f(t) \cos\left(\frac{n\pi t}{L}\right) dt$$

$$= \frac{1}{4} \int_{-4}^4 f(t) \cos\left(\frac{n\pi t}{4}\right) dt$$

$$= \frac{1}{4} \left(\int_{-4}^0 -\cos\left(\frac{n\pi t}{4}\right) dt + \int_0^4 \cos\left(\frac{n\pi t}{4}\right) dt \right)$$

$$= \dots = 0 + 0 = 0$$

no cosine terms at all

$$b_n = \frac{1}{4} \int_{-4}^4 f(t) \sin\left(\frac{n\pi t}{L}\right) dt$$

$$= \frac{1}{4} \left(\int_{-4}^0 -\sin\left(\frac{n\pi t}{L}\right) dt + \int_0^4 \sin\left(\frac{n\pi t}{L}\right) dt \right)$$

$$= \dots = \frac{2}{n\pi} [1 - \cos(n\pi)] = \frac{2}{n\pi} (1 - (-1)^n)$$

$$= \begin{cases} \frac{4}{n\pi} & \text{if } n \text{ is odd} \\ 0 & \text{if } n \text{ is even} \end{cases}$$

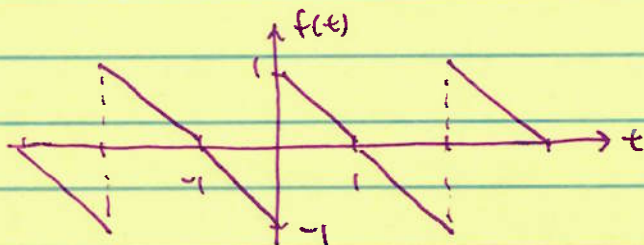
$$\text{So, } f(t) \sim \sum_{n=1}^{\infty} \frac{2}{n\pi} (1 - (-1)^n) \sin\left(\frac{n\pi t}{4}\right)$$

$$\sim \sum_{n \text{ odd}} \frac{4}{n\pi} \sin\left(\frac{n\pi t}{4}\right)$$

$$\sim \frac{4}{\pi} \sin\left(\frac{\pi t}{4}\right) + \frac{4}{3\pi} \sin\left(\frac{3\pi t}{4}\right) + \frac{4}{5\pi} \sin\left(\frac{5\pi t}{4}\right) + \dots$$

show graph

another example: $f(t) = \begin{cases} -1-t & -1 < t < 0 \\ 1-t & 0 < t < 1 \end{cases}$ period 2 ($L=1$)



$$a_0 = \frac{1}{2} \int_{-1}^1 f(t) dt = 0 \quad (\text{net area is } 0)$$

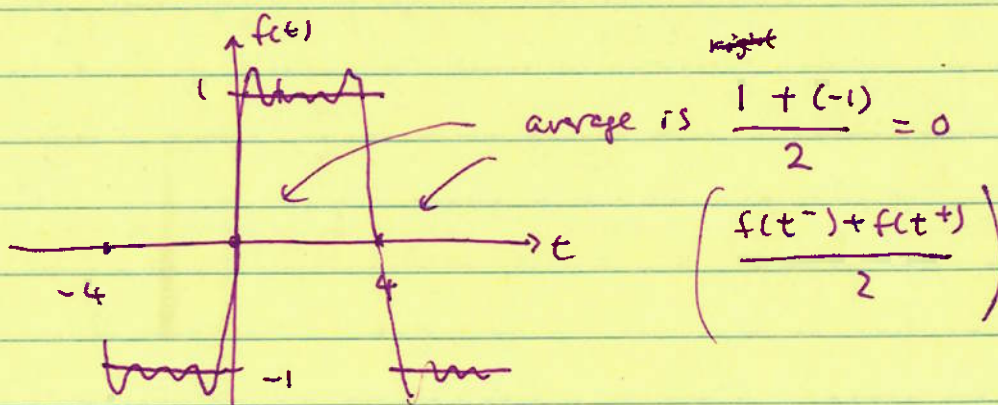
$$a_n = \frac{1}{2} \int_{-1}^1 f(t) \cos\left(\frac{n\pi t}{1}\right) dt = \dots = 0$$

$$b_n = \int_{-1}^1 f(t) \sin(n\pi t) dt = \dots = \frac{2}{n\pi}$$

$$\text{so, } f(t) \sim \sum_{n=1}^{\infty} \frac{2}{n\pi} \sin(n\pi t)$$

$$\sim \frac{2}{\pi} \sin(\pi t) + \frac{2}{2\pi} \sin(2\pi t) + \frac{2}{3\pi} \sin(3\pi t) + \dots$$

notice the common features: when there is a discontinuity, the Fourier series goes through the middle (average value)



why? easy to see for these two:

$$\text{first one is } \frac{4}{\pi} \sin\left(\frac{\pi t}{4}\right)$$

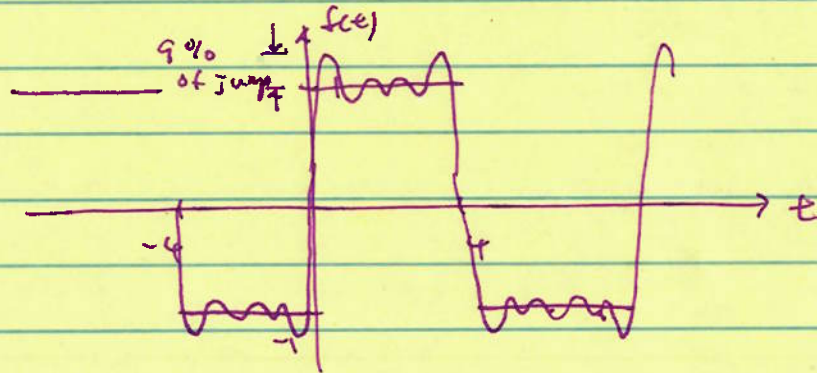
$$\sum_{n=1}^{\infty} \frac{2}{n\pi} (1 - (-1)^n) \sin\left(\frac{n\pi t}{4}\right)$$

discontinuity is at $t=0$ and all

$\sin\left(\frac{n\pi t}{4}\right)$ are equal to 0 so the

Fourier series has no choice but to go 0

2nd common feature: the overshoot (Gibbs phenomenon)



the overshoot is always around 9% of the jump, regardless of the order (moves closer to discontinuity as n increases)

→ harshness of "8-bit" sounds